Definition of Terms Transportation Engineering

Alligator cracking – a series of interconnected of interfaced cracks caused by fatigue failure of the asphalt concrete surface under repeated traffic loading

Amplitude - the height between the trough and the crest of a wave

Asphatic Concrete – a layer consisting of properly designed mix od coarse aggregate fine aggregate, a filler and a bituminous binder

Average speed – the average of the spot speeds of all vehicles passing a given point in the highway

Barriers – highway appurtenances designed to prevent vehicular penetration from the travel way to areas behind the barrier such as to minimize damage to impacting vehicles and their occupants and to reduce the risk of injuries to pedestrians and workers

Base - the component of the road structure which receives the traffic load and transfer it to the subgrade

Base courses - this normally consists of aggregates such as gravel and crushed rock

Basic capacity - the max number of passenger cars that can pads a given point on a lane or roadway during one hour under the most nearly ideal roadway and traffic conditions which can possible be attained

Batching – process of proportioning cement, water, aggregates and additives prior to mixing concrete

Bathymetry – the physical configuration of the seabed, the measurements of depths of water in the ocean, etc. and also information derived from such measurement

Beuforts Scale – an instrument use to measure intensity of wind

Bitt - the mooring fitting for mooring ships during a storm installed close to the water line of the berth

Bitumen carpet – a 20 mm thick premix bitumen layer over which is laid a seal coat

Bleeding or flushing – the exuding of bitumen onto the pavement surface causing a reduction in skid resistance

Block cracking – cracks forming large interconnected polygons, usually with sharp corners or angles. These cracks are generally caused by hardening and shrinkage of the asphalt and/or reflection cracking from underlying layers such as cement treated base

Blowups - localized upward buckling and shattering of the slabs at traverse joins or cracks

Breakwater – the structure that protects harbor from stormy waves and permits calm in the harbor

Braking distance – distance traveled by the vehicle after the application of the brakes until it will stop

Breaking waves – waves which fall forward since the forward velocity of the crest particles exceeds the velocity of propagation of the wave itself. In deep water, this normally occurs when the wave length L is less than 7 times the wave height H ($L < 7H$) and in shallow water when the depth d is approximately equal to 1.25 H

Breasting dolphins - Type of dolphins which are designed to take the impact of the ship when docking and to hold the ship against a broadside wind. Therefore, they are provided with fenders to absorbed the impact of the ship and to protect the dolphin and the ship from damage.

Bollard - a vertical post to which the eye of a morring line can be attached

- the mooring fitting for mooring ships during a storm installed at the outside or outside/inside of the berth far from the waterline

Broken back curve - these consist of two curves in the same direction separated by a short tangent

Bulkhead – a structure for retaining or to prevent earth or fill from sliding into water

Capacity - the maximum sustained 15 min rate of flow, expressed in passenger cars per hour per lane, which can be accommodated by a uniform freeway segment under prevailing traffic and roadway conditions in one direction

Catch point - the points at which the cut or fill slopes intersect the existing ground

Chevron markings – markings that are often used to guide traffic into the right turning

lanes separated by an island, such as corner island at a signalized intersection

Circular curves - these are usually laid out in the field by occupying the tangent to curve point TC with a transit and then establishing successive points by turning deflection angles and measuring chords

Clapotis – the pressure against a vertical wall due to waves

Concrete - this term refers to any material consisting of a mixture of aggregates, such as sand, gravel, or crushed stone, fastened together by cement

Compound curve - these result when two curves of differing radius join one another

Construction joint – joints provided in concrete cement pavement is suspended

Continously reinforced concrete pavements – a type of rigid pavement, which has no transverse joints, except construction joints or expansion joints when they are necessary at specific positions such as at bridges. This type of pavement is typically used on high-volume, high-speed roadways

Cost of borrow – this is the cost of any material that has to be obtained off job site in order to make a fill

Crossing conflicts - this traffic conflict occurs when they cross paths directly

Cross slope - these are normally specified in m/m or in percentages. In railway practice, cross slopes may be expressed in millimeters that the high rail is above the low rail, since there is a standard horizontal distance maintained between rails

Curing – the process of providing the required water and maintaining a favorable temperature for a period of time after placing concrete

Cut or excavation - the existing ground is above the construction section

Drainage – primary consideration in the design of geometric cross sections for highways, runways, and taxi ways

Deep water waves – waves in which the ratio of water depth to the wavelength is greater than 0.5

Delay – difference between the actual travel time and a given segment of a transportation system and some ideal travel time for that segment

Delineator – light retro reflecting devices mounted at the side of the roadway in series to indicate the roadway alignment

Density – it is defined as the number of vehicles per unit distance occupying a thsection of roadway at a given instant time and is usually measured in vehicles per mile or per km

Design speed - defines as the maximum safe speed that can be maintained over a specified section of highway when conditions are so favorable that the design features of highway govern

Diagonal markings – markings which are placed on sealed shoulders or other sealed portion of the road where traffic is not desired.

Diffraction – the spread of energy along a wave crest

Distance gaps – defined as the fraction of time that vehicles are present at a point in space

Distance separation - the reciprocal of density which is usually measure in vehicles per mile or per kilometer

Diurnal tides - tide which occurs only one high tide a day

Diverging – the process by which a vehicle in a traffic stream leaves that traffic stream, such as a vehicle leaving the outside lane if an expressway

Diverging conflicts - this traffic conflict occurs when vehicles leave the traffic stream

Drainage - the primary consideration in the design of geometric cross sections for highways, runaways, and taxiways

Drip track raveling – progressive disintegration of the surface between the heel paths caused by dripping of a gasoline or oil from vehicles

Dry dock - an artificial basin for vessels when the basin is pumped out

Duration – the time that a wind blows across the water

Earthwork - these quantities are normally expressed as volumes, in metric units, they are given in cubic meters

Elevation - these are in meters above some datum; usually this datum is mean sea level, but it may be any arbitrary scale

Emotion – the time linked with the two stages perception and intellection based on a particular traffic situation like fear or anger which has a vital influence on the final message or decision sent by the brain to the muscle

Estimates - these includes cost estimates for various parts of the project and are used too evaluate the acceptability of bids and financial feasibility of the project

Expressway – divided arterial highway for through traffic with full or partial control of access and generally with grade separations at major intersection

Facility oriented organization - primarily involved in planning designing, constructing, maintaining, and operating fixed facilities

Fairway – an open water with navigable depth

Faulting - elevation difference between adjacent slabs at traverse joints. This is usually the result of pumping and is a major source of Portland concrete pavement

Fetch – the distance that the wind blows over the sea in generating the waves is known as:

Fill or embankment - the existing ground is above the construction section

Fixed mooring berth - A marine structure consisting of dolphins for tying up the ship and a platform for supporting the cargo handling equipment.

Flashing lamps - warning devices use to supplement either controls and devices necessary to alert motorist of construction and maintenance activities of obstruction in roadway

Flexible - a pavement which has negligible flexural strength

Flow - defines as the number of vehicles per unit time passing a point in space and is usually expressed in vehicles per hour

Free haul distance - the cost of excavation normally includes the cost of removing the material from its existing state, hauling it up to some specified distance

Gap – Th headway in a major stream, which is evaluated by a vehicle driver in a minor stream who wishes to merge into the major stream

Geoemtric cross section - this view has elevation as its vertical axis and horizontal distance, measured perpendicular to the centerline, as its horizontal axis

Geotextiles - these are sometimes used in constructing pavements. These are used to add strength, control moisture, and prevent the movement of fine materials into drainage layers

Grade - expressed as decima fractions or as percentages

Grade separation - these traffic solutions eliminates the crossing conflicts by placing the conflicting traffic streams at different elevations at their point of intersection

Gravity waves - Waves are formed by the frictional drag of wind across the water surface. This is a process of transferring energy from wind to water. Water particles are moved from their position by the wind, and then returned to the original position by gravity, which is a restoring force.

Groin – a shore protection structure usually built perpendicular to the coastline to retard littoral transport of sedimentary materials

Guide sign – inform and advice road users of directions, distance, routes, the location of services and points of interests

Haul - the product of a volume of materials times the average distance it is hauled

Headway – the reciprocal of flow is the average time separation in the traffic stream and is usually measured in seconds

Highest wave - A maximum wave height and wave period of the maximum wave height in wave train.

Horizontal Control – this type of control on the object on the job site is commonly identified as primary, secondary and working

Horizontal tangents - these are described in terms of their lengths and their directions

Immediate water waves – waves in which the ratio of water depth to the wave length is less than 0.5 but greater than 0.40

Intellection - after perception occurs, the formation of new thoughts and ideas, recalling old memories of similar occasions to a particular traffic situation

Jetty - a solid structure, which projects into the sea perpendicular to the shore to berth vessels

Joint or crack spailling – the breakdown or disintegration of slab edges at joints or cracks, usually resulting in the ;ss of sound concrete and the progressive widening of the joint or crack

Lag distance - distanced traveled by the vehicle during the total reaction time

Leeward side - the opposite side of a structure facing the direction from which the wind comes

Limit of economic haul - the longest distance material should ever be hauled

Longitudinal cracking – cracks approximately at right angles to the pavement center line. These are caused by poorly constructed construction joints and shrinkage of the asphalt concrete surface; they may also be reflective cracks

Mass diagram - calculation of optimum haul strategies and earthwork costs is done

Mastic asphalt - a mixture of bitumen, fine aggregate and filler in suitable proportion heated to about 200 C in special cooker and laid

Maximum free speed – the maximum speed that cab be attained on a highway as the flow tends to zero

Mean free speed - the maximum speed that can be attained on a highway as the flow tend to be zero

Merging – the process by which a vehicle in on traffic stream joins another traffic steam moving in the same direction such as a ramp vehicle joining a freeway stream

Merging conflicts - this traffic occurs when vehicles enter a traffic stream

Macroscopic - these models of traffic flow describe the relationship among flow, speed and density

Microscopic - these models of traffic flow describe the motion of individual vehicles and their interactions with one another

Mixed diurnal tides - if one of the two daily tide is incomplete that is it does not reach the height of the previous tide, then the tides

Mole – A breakwater is a structure constructed for the purpose of forming an artificial harbor with a water area so protected from the effect of sea waves as to provide safe shipping. When a accommodations for breakwater supports a roadway

Moonson - prevailing winds which are seasonal blowing in one direction over part of the year and in the opposite direction the remainder of the year

Mooring dolphins - type of dolphins which are not designed for the impact of the ship, as they located in back of the face of the dock where they will not be hit. They are located about 45 off the bow and stern.

Neap tides – the lowest tide of the month when the line connecting the earth with sun and the moon form a right triangle that is when the moon is in its quarters

Occupancy - defined as the fraction of time that vehicles are present at a point in space

Operating organization - known as carriers, primarily concerned with operating fleets to provide transportation services

Overhead sign – signs which provide means of displaying essential traffic information on wide multilane roads, where some degree of lane use control is required or where side of road clearance is insufficient to accommodate a road side sign

Parkway – an arterial highway for non-commercial traffic with full or partial control of access and usually located within park

Passing sight distance – minimum sight distance required on a two lane, wo way highway that will permit a driver to complete a passing maneuver without colliding with an opposing vehicle and without cutting off the passed vehicle

Pavement - the layered structure placed over a soil sub grade for forming a road

Pavement deflection – refers to the structural adequacy of the pavement section

Pavement distress – refers to the condition of a pavement in terms of its general appearance

Pavement roughness – the measurement of the extent to which a road surface deviates from the plane

Perception time - time required for transmission of the sensations received (through eyes, ears and body) to the brain and the spiral chord by the nervous system, to a particular traffic situation

Period - the interval of time for two wave crests to pass the same position in space

Pier – a berth structure projecting out from the shoreline

Plans - these refer to drawings, usually accompanied by notes, of various aspects or components of the design

Plain concrete pavement – a type of rigid pavement, which has no temperature steel or dowels for the load transfer. This type of pavement is used mainly on low volume highways or when cement stabilized soils are used as sub material

Plain view - this is a drawing of the facility as it would look to an bserver directly above it

Profile - this view has elevation as its vertical axis and horizontal distance, measured perpendicular to the centerline, as its horizontal axis

Profile grade - line representing the facility on the profile

Port - a sheltered place where the ship may receive or discharge cargo. It includes the harbor with its approach channels and anchorage places

Possible capacity - the max number of passenger cars that can pass a given point on a lane or roadway during one hour under prevailing roadway and traffic conditions

Practical capacity - the max number of vehicles that can pads a given point on a lane or roadway during one hour without traffic density being so great as to cause unreasonable delay hazard or restrictions to the driver's freedom to maneuver under the prevailing roadway and traffic

Prime coat - a layer of stone chipping coat laid over a hot to make the surface water proof conditions

Quay - a platform built in the harbor parallel to the shore and backed up by ground

Queue - a delay which results when the demand for a facility exceed its capacity

Raveling – wearing away of the pavement surface caused by the dislodging of aggregate particles and binder

Reaction time - the total time required for perception, intellection, emotion and volition that is from the instant the object comes in the line of sight of the driver to the instant he arrives a decision say to slow down or overtake, under normal circumstance

Refraction - the bending of waves as they slow down

Regulatory signs – signs that inform road users of traffic laws and regulations which, if disregarded, will constitute an offense

Releiving platform - a platform or deck structure built below the top deck level and supported on bearing piles. The main function of the platform is to reduce the lateral soil pressure over the upper portion of the sheet wall

Roadwork signs – warn or advise if temporary hazardous conditions that could endanger road users or the men and equipment engaged in roadwork

Road capacity - the maximum number of passenger cars that can pass a given point on lane or roadway during one hour under the most nearly ideal roadway and traffic conditions which can possibly be attained

Rumble strips – type of thermoplastic Lane marking designed to aid and provide motorist with visual, audio and motion warnings on the road

Running speed – the average speed maintained by a vehicle over a particular stretched of road, while the vehicle is in motion. This is obtained by dividing the distance covered by the time during which the vehicle is in motion

Screeding – to level a floor or layer of concrete with a straight edge using a back and forth motion while moving across the surface

Sea - waves under the influence of the winds that generated them

Sea island - a berth structure with no direct connection to the shore, at which the ships can berth. Berthing can take place on either one or both sided of the structure

Seal coat – a very thin surface applied over a bituminous pavement to make it impervious

Seiche – a very long-standing wave on a large but limited body of water (lake) generated by occurring when a storm dies down after producing a wind tide

Seiching waves - waves of this type have very long periods, typically from 30 sec. up to the tidal period of 12 hr. 25 min. and are mostly found in enclosed or semi-enclosed basins such as artificial port basins or bays.

Semi diurnal tides – tides that occur twice its lunar day

Settlement - local sagging in the pavement caused by differential settlement, consolidation, or movement of the underlying earth mass

Shallow water waves – waves in which the ratio of water depth to the wave length is equal to or less than 0.40.

- waves which occur in water having a depth less than one half of the wave length and the influence of the bottom changes the form or orbital motion from circular to elliptical or near-elliptical.

Shoulders - these are intended primarily as a safety feature. They provide for accommodation of stopped vehicles, emergency use, and lateral support of the pavement

Skid resistance – describes the effectiveness of a pavement to prevent or reduce skid related crashes

Significant wave - A hypothetical wave having a wave height and period equal respectively to average values of the wave height and period of the largest 1/3 of all waves in the train as counted in the order of greater wave height.

Simply reinforced concrete pavement – a type of rigid pavement having dowels for the transfer of traffic load across joints with these joints spaced at larger distance than with plain pavements ranging from 10 m to 30 m. Temperature steel is used throughout the slab

Single and double pillar - A type of mooring bollard which is suitable for warping ships along berths a single pillar type is used with lines from one ship only. General mooring application where rope angle is not steep. It has a normal max. working capacity of 200 tons.

Significant wave height –the arithmetical mean value of the highest one-third of the waves for a stated interval

Slobing robe - A type of mooring bollard where all general mooring applications including steep rope angles and lines from two ships may be attached without interference. It has normal max. capacity of 200 tons.

Swell - waves that have propagated beyond the initial winds that generated them

Swells Waves – generated by storms, which occur outside area of observation

Space headway – the distance between the front of a vehicle and the front of the following vehicle

Space mean speed – the arithmetic mean of speed of vehicles occupying a relatively long section of street or highway at given instant

Special instruction signs – signs that instruct road users to meet certain traffic rule requirements or road condition

Specifications - these are written instructions detailing how the facility is to be constructed

Speed - rate of change of distance with respect to time and is usually measured in either miles per hour or feet per second depending on the application

Spot speed - the instantaneous speed of a vehicle at a specified section or location

Spring tides – the highest tide which occurs at intervals of half a lunar month when the sun , moon and earth fall in line

Steepness – the ratio of the wave height to its wave length is called

Stopping sight distance – minimum sight distance required for a driver to stop after seeing an object in the vehicle's path without hitting that object

Storm Surge – an abnormal rise of the sea level that occurs when a typhoon passes by. This rise above normal level on this open coast is due to atmospheric pressure reduction as well as due to wind stress.

Structural number - determined based on the traffic loads, soils support, and other design factors

Sub base – the part of the road structure which is immediately above the subgrade and composed of stone boulders or superior soil

Sub grade - the layer of natural soil over which the pavement road is laid

Superelevation diagram - this applies to cuved facilities, such as highways or railways, only. It consists of a graph with roadway or railway cross slope versus horizontal distance

Super elevation run off - the length of super elevation development from zero cross slope to full design super elevation in a circular curve alignment

Swell waves - Wind generated waves which are created in the deep ocean at some distance from the port site and the wind that created them may be too distant to be felt in the port or may have stopped blowing or changed, its direction by the time the wave reach the port.

Superelevation or banking of curves - the purpose of this is to counteract the centripetal acceleration produced as a vehicle round a curve

Superelevation transition - this involve modification of the roadway cross section from normal crown to full superelevation, at which point the entire roadway width has a cross slope of e

Surface attrition or surface abrasion - abnormal surface wear, usually resulting from poor quality surface mortar or coarse aggregate

Surface polish – loss pf the original surface texture due to traffic action

Tack coat - an application of hot bitumen material given to the old surface to provide adhesion to the old and new road surface

Tangent run out - the length of super elevation development from the normal cross slope to the zero cross slope point on the tangent

Tee head - A type of mooring bollard where all general mooring applications including steep rope angles. Any one bollard should preferably be allocated to lines from one ship only. It has a normal maximum working capacity of 150 tons.

Temporary curbing – roadwork devices consisting of pre cast concrete sections, sandbag, and other which, may be used to guide traffic at the construction site

Tidal bores - high crester waves caused by the rush of flood tide up river or by meeting of tides

Tide - waves formed by gravitational attraction of the moon and sun

Time headway – the time interval between the passage of successive vehicle is moving in the same lane and measured from head to head as they pass a point on the road

Time mean speed – the arithmetic mean of the speeds of vehicles passing a point on a highway during an interval of time

Time sharing - these traffic solution involve assignment of the right of way to particular movements for particular time

Transitional water wave – a type of wave where depth is less than ½ of wave length but greater than 1/25 of the wave length

Traffic capacity – the ability of a roadway to accommodate traffic volume. t is expressed as the maximum number of vehicles in a lane or a road that can pass a given point in unit time, usually an hour, that is vehicle per hour per lane or roadway

Traffic cone – devices which may be conical in shape or tubular shaped capable of performing channelization o traffic which may be set on the surface if the roadway or rigidly attached for continued use

Traffic congestion - a major challenge for the transportation system and the transportation engineering profession because it undermines the effectiveness and efficiency of the system

- demand exceeds the capacity of the transportation system

Traffic signs – device mounted on a fixed support or portable support whereby a specific message is conveyed by means of words or symbols placed or erected for the purpose of regulating warning or guiding taffic

Traffic safety - continuing challenge for the transportation engineering profession

because of public expectations that safety will continue to improve.

Traffic volume – the number of vehicles moving in a specified direction on a given lane or roadway that pass a given point during specified unit time and is usually expressed as vehicles per hour or vehicle per day

Transition curves - these are used to connect tangents to circular curves

Traverse cracking – cracks approximately at right angles to the pavement center-line. These may be caused by shrinkage or differential thermal stress of the asphalt concrete, or may be reflective crack

Traverse joint – joints provided in concrete cement pavements at right angles to the centre line of the pavement

Tsunamis – waves created by large, sudden impacts, such as earthquakes, volcanoes or landslides that ends up in ocean.

Vehicle dynamic - this determines the length of the superelevation run off of transitional curves

Velocity of wave propagation – the speed pf the wave form which travels over the water surface for a certain wave

Vertical alignment - the longitudinal profile along the centerline of the road. It is made up of a series of grades and vertical curves

Vertical curve - these are normally parabolas centered about the point of intersection of the vertical tangents they join

Volition time - the time taking a decision to produce action to a particular traffic situation

Volume - number of vehicles counted in a specified time interval

-maximum number of vehicles, passengers or the like, which can be accommodated by a given facility or system under given conditions at a given level of service

Wakes – waves formed by moving ship or boats

- waves formed by moving ship or boats

Warning signs – warn road users of condition on or adjacent to the road that may be unexpected or hazardous

Water bound macadam - a road made of coarse aggregate mechanically interlocked by rolling and bonded together with screening dust

Wave Crest - the highest point of the wave

Wave height - the elevation of the crest above the through of the wave

Wave length - the length between two consecutive crests

Wave period – the wave form travels over thee water surface and the time for two consecutive crests to pass a point

Wave refraction - A change in the dissection of travel of the wave with change in depth of water which distributes wave along energy the seashore unevenly

Wave trough - the lowest point of the wave

Wearing coarse - the layer in a road pavement which provides resistance to wear and tear due to traffic

Wearing Surface - soil stabilized roads with cement/lime cannot be used

Weaving – the process by which a vehicle first merges into a stream of traffic obliquely crosses that stream, then merges in to a second stream moving in the same direction

Weaving conflicts - this traffic conflict occurs when vehicles cross paths by first merging and the diverging

Wharf - a platform built parallel to the shore or breakwater within the harbor to berth vessel

Wind - the circulation of masses of air more or less parallel to the earth's surface

Wind rose - the graphical representation of the direction, frequency and intensity of winds at a particular location over period of time

Wind Seiching – when local winds blow over the surface of the water, near the shores will create

Windward side – the side of a structure facing the direction from which the wind

Wind waves - waves generated by wind that are acting on the sea surface bordering on the port site.

Definition of Terms Fluid Mechanics & Hydraulics

Absolute Pressure – pressure above absolute zero

Archimedes Principle- Anybody immersed in a fluid is subjected to an upward force called buoyant force equal to the weight of the displaced fluid.

Atmospheric pressure – pressure at any one point on the earth surface from the weight of the air above it.

Bernoulli - in any stream flowing steadily without friction the total energy contained is the same at every point in its path of flow

Buttress Dam - consist of wall, or face, supported by several buttresses on the downstream side

Dam – structures that block the flow of a river, stream, or other waterway

Drag Force – It acts opposite to the direction of the oncoming flow velocity

Discharge – the volume of liquid passing a cross section of a stream

Embankment Dam - gravity dam formed out of loose rock, earth, or a combination of these materials

Froude Number – It is a dimensionless value that describes different flow regimes of open channel flow

Gates – an opening in a dam or other hydraulic structure to control the passage of water

Gravity Dam - used only the force of gravity to resist water pressure

Hydraulic grade line – graphical presentation pf the total energy of flow

Hydraulic Gradient – a line joining the points of highest elevation of water of a series of vertical open pipes rising from a pipeline in which water flows under pressure

Hydraulic Jump – abrupt increase in depth of rapidly flowing water

Hydrodynamics – deals with the study pf the motion of liquid and of the interactions of the fluid with its boundaries

Laminar Flow – when the path of individual fluid particles do not cross or intersect

Major head loss - caused by pipe friction along straight sections of pipe of uniform diameter and uniform roughness

Metacenter – the point of intersection between the axis of the body and the line of action of the buoyant force

Metacentric height – the distance from the metacenter to the center gravity of the body

Minor head loss – caused by changes in velocity or directions of flow an are commonly express in terms of kinetic energy

Nozzle – a converging tube installed at the end of a pipe or hose for the purpose of increasing the velocity of the issuing jet

Orifice – an opening with a closed perimeter through which fluid flows

Pascal - the pressure exerted onto a liquid a transmitted undiminished to all portions of the liquid

Pitot Tube – used to measure the velocity of fluid flow or velocity of air flow as used in airplane speedometer

Power – rate of work done per unit of time

Relative Pressure – pressure above or below the atmosphere and can be measure by pressure gauges or manometers

Reynolds Number – defines as the ration of inertia force to viscous force and interpreted as the ratio dynamic pressure to shearing stress

Turbine – extract flow energy to do mechanical work which in turn converted into electrical energy for turbines

Turbulent Flow - when the path of individual particles are irregular and continuously cross each other

Viscosity – resisting property of a liquid to shearing force

Water hammer - a pressure surge or wave caused when a fluid in motion is forced to stop or change direction suddenly

Weir – overflow structures which are built across are open channel for the purpose of measuring or controlling the flow of liquids

Venturi Meter – an instrument used in measuring the discharge through pipes



Engineereya's Vlog

Definition of Terms Geotechnical Engineering

Active earth pressure coefficient – ratio between lateral and vertical principal effective stresses when an earth retaining structure moves away from a retained soil

Allowable bearing capacity or safe bearing capacity -the working pressure that would ensure a margin of safety against collapse of the re from shear failure.

Cohesion – the component of shear strength of a rock or soil that interparticle friction

Consistency – used to describe the degree of firmness of soil

Embedment depth (D) - the depth below the ground surface where the base of the foundation rests

Equipotential line - a line along which the potential head at all points is equal

Factor of safety or safety factor - the ratio of the ultimate net bearing to the allowable bearing cap ng capacity or to the applied maximum vertical stress

Flow Line - a line along which a water particle will travel from upstream to the down stream side in the permeable soil medium

Flow Net - a combination of a number of flow line and equipotential line

Footings - consisting of a small slab for transmitting the re load to the underlying soil.

Foundation - structure load to is that part of a structure which transmits the building, load nto the underlying soil

Intergranular Stress – stress resulting from particle to particle contact of soil

Liquefaction – a process by which water saturated soil sediment temporarily loss strength and acts as a fluid

Overburden pressure - the pressure (effective stress) of the soil removed fraction of the to place the footing

Over consolidated – it is a soil whose present effective over burn pressure is less than that which the soil experienced in the past

Passive earth pressure coefficient – ratio between lateral and vertical principal effective stresses when an earth retaining structure is forced against a soil mass

Pore water pressure – stress induced by water pressures

Retaining wall – structure whose primary purpose is to prevent lateral movement of earth or some material

Shear – soil derives its strength from its capacity to resist

Total Stress – sum of effective and neutral stresses

Shallow foundation - one in which the ratio of the embedment depth to the minimum plan dimension, which is usually the width, is Di/ B 52.5.

Ultimate bearing capacity - the maximum pressure that the soil can support

Ultimate net bearing capacity - the maximum pressure that the soil can support above its current

Definition of Terms Structural Engineering Construction

Actual displacement - it is measured by seismometer

Center of rigidity – it is the point through which the resultant of the resistance to the applied lateral force acts

Center of mass – it is the point through which the applied seismic force acts

Column – structural member ha the ratio of its unsupported height to its least lateral dimension of not less than 3 and is used primarily support axial load

Corbel – a short edge beam projecting from a column to support a weight

Creep – the structure is subjected to a load for a long period of time, it will continue to deform until a sudden fracture occurs

Ductility – it refers to the ability of a material to deform in the plastic range without braking

Elasticity – it refers to the property of a material which makes it return to its original dimension when the load is removed

Eccentricity – it is the distance between the center of rigidity and center of mass

Fatigue – the structure is subjected to a cycle of stresses and _____, it causes the beam to have sudden ultimate structure

Focal depth – besides the epicenter, it describes the location of the earthquake

Homogenous – the material has the same composition at every point but the elastic may not be the same in all direction

Hookes Law – stress is proportional to the strain within the elastic region

Isotropic - The material has the same elastic properties in all direction

Kinetic Friction – retarding force acting opposite in motion

Magnitude of earthquake – it is measured by the Ritcher scale

Orthotropic – The composite material exhibits elastic properties in one direction different from that in the perpendicular direction

Poisson's Ratio – the deformation of axially loaded members, the ratio of the lateral to the longitudinal strain is constant

Post tensioning – stressing high strength steel after the concrete has been cast and has attained sufficient strength

Pre tensioning – stressing high strength steel wires before concrete hardens

Plasticity – A condition that when there is already a permanent deformation, it continues to deform when a minimal load is applied beyond the elastic

Reciprocal of deflection – it refers to the rigidity of a structure

Reciprocal of stiffness – it refers to flexibility of a structure

Relaxation –Loss of stress that takes place with the passage of time as concrete is held at a constant strain

Resilience – it refers to the ability of a material to absorb energy in the elatsic range

Resonance – Refers to the large amplitude vibration of an object or system when given impulses at its natural frequency

Soft Story – ne in which the lateral stiffness is less than 70 percent of the stiffness of the story above

Resultant – determines whether the body will be in equilibrium or will have a varying state of motion

Static it refers to the force generated by a body at rest

Story Drift – it is the displacement of one level relative to the level above or below

Torsional Shear Stress – occurs when the center of mass and rigidity do not coincide

Yielding – the material deforms considerably even with a slight increase in stress

Toughness – it refers to the ability of a material to absorb energy in the plastic range

Young's Modulus – it is the constant of proportionality that defines the linear relationship between stress and strain



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